Multiple connections need one robot owner.

The original transport accepted multiple clients; shared interaction authority was incomplete.

Before / independent intent

Each headset could submit motion.
Drafts and modes lived locally.
Latest client pose could affect agent.
One slow receiver could lose controls.

Now / authoritative state

One robot-wide lease, one epoch.
Server validates motion and revisions.
Shared targets, preview, mode, plans.
Every client may Emergency Stop.

Native robot execution

Navigation + optional arm adapter.
Accepted operations outlive edits.
Late work is fenced before effects.
Agent/manual needs quiescence.

Explicit replication boundaries

ARModule / replication

Authority, leases, epochs
Canonical target snapshots

ClientCore / replication

Prediction + interpolation
Shared portable interaction facts

Both hosts / replication

Grey cross + input eligibility
Apply each headset's own origin

Scope: one robot, mixed Specs/WebXR peers; no upstream DimOS edits or scene-file edits.

Fast for the owner. Smooth for observers.

Local prediction is immediate; the server remains the source of accepted robot state.

Client A / dragging

Render local pose immediately.
Acquire / renew robot-wide lease.
Send fenced target samples.

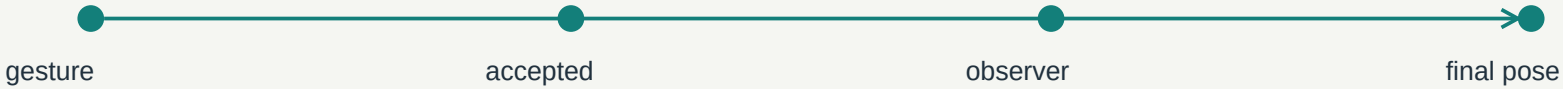
ARModule / canonical odom

Validate epoch + lease + sequence.
Publish shared state at up to 20 Hz.
Keep nav and arm groups distinct.

Client B / observing

Buffer samples for about 100 ms.
Interpolate position + rotation.
Render using B's tracking origin.

Replica timing



Owner prediction never waits for the network. Native commands wait for authority.

Observers do not extrapolate. Quaternion interpolation takes the shortest rotation path.
Final samples settle exactly; stale / reordered snapshots cannot revive an old gesture.
Reliable controls preserve order. Coalesced telemetry avoids unbounded backlog; failed peers retire.
Late join and reconnect rebuild from full state; parked targets remain available for handover.

Only one client plans the selected target.

The reservation survives the first release so the two-gesture preview workflow stays exclusive.

1 / First gesture

A grabs valid floor target.
Robot-wide lease granted.

2 / First release

Shared preview remains.
A keeps planning reservation.

3 / Second gesture

A sets heading and confirms.
Server accepts native goal.

4 / Handover

Reservation ends.
B can grab shared target.

Observer B / unavailable

Sees A's target and advisory route.
Grey marker with cross; grabbing is disabled.
Emergency Stop and cancellation controls stay reachable.

Reservation cleanup

Confirm / cancel / deselect / tracking loss.
Disconnect or lease expiry also retires authority.
Normal live release opens authority and keeps the target.

A 500 ms lease is renewed during the reservation. Losing renewal cannot leave an immortal lock.
Arm execution has a separate robot-owned busy state; ending an edit does not unlock an active operation.

One world, independently aligned headsets.

AprilTag math is reusable across clients; calibration and capture timing decide physical accuracy.

Robot side

Native odometry
Measured base-to-tag mount

Observed tag

Known black-square size
Camera optical pose at exposure

Headset side

Per-client tracking origin
Never shared between headsets

T_odom_client = (T_odom_base x T_base_tag) x inverse(T_client_camera x T_camera_tag)

Implemented corrections

Keep existing bracket interpolation acceptance.
100 ms applies only to nearest-sample fallback.
Keep canonical targets in odom; reproject on correction.
Reference resets and capture IDs fence stale work.

Physical checks still required

Verify tag size and rigid robot mount by measurement.
Validate camera intrinsics, crop and extrinsics per device.
Check two-headset target overlap while both move.
Wrong size or mount can still give low reprojection error.

Synthetic tag/PnP checks and unit tests support the transform chain; they do not certify device alignment.
Built-in Quest calibration assumptions still need measured-device verification.

Keep bulk data away from control.

Compact state, cached geometry and bounded workers protect interaction responsiveness.

Control channel

Authority, Stop, targets and replies.
Small revision references, not full plans.
Bounded reliable queue; snapshots coalesce.
Owner input renders without an ACK wait.

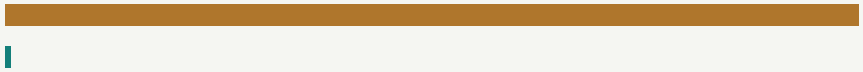
Paired bulk channel

Model / route / trajectory by revision.
JPEG uploads and latest-only LiDAR.
Small records and incremental decoding.
Download budget; timely image uploads.

Bounded native work

Latest-only sensor and costmap inputs.
Control stays off perception worker lanes.
Hung work cannot spawn replacements.
Late results are generation-fenced.

Arm-state bytes / update: synthetic 6 joints, 1,000 trajectory points



Release frees editing; the server finishes motion.

Takeoverable finishing

Continue toward the last validated arm target.
No former-client 30 Hz finishing stream.
A new grab replaces the native motion generation.
Server loss, Stop or last disconnect ends the tail.

Preserved presentation

Observers use the existing 100 ms target buffer.
Measured robot feedback remains separate from intent.
Cache model/path geometry; update transforms per frame.
Execute and reset retain exclusive operation ownership.

Five peers at 10 Hz: arm-state application traffic falls from 7.47 MB/s to 41.95 KB/s in this workload.
Loopback control ACK p95: about 2.5 ms during bulk transfer. Synthetic Mac evidence; not Wi-Fi or headset results.

Test the races, then verify on devices.

Automated coverage exercises shared state and native boundaries; hardware remains a separate acceptance step.

Unit / deterministic clocks

Competing grabs; delayed grants.
Lease expiry; renew; cancellation.
Epoch / sequence / revision fences.
Predicted and interpolated targets.
Native plan / execute late results.
Exposure timestamps and alignment.
Late assets / delayed disconnect cleanup.

Integration / production owners

Real multi-socket ARModule tests.
Five browser contexts, one server.
Shared preview and grey cross.
Confirm / handover / disconnect.
Native ports simulated explicitly.
Native Specs compiler checked.
Release / accepted-motion races.

Device / acceptance

Specs + Quest concurrent session.
Moving tags, occlusion, relocalize.
Wi-Fi jitter and temporary loss.
Observed drag smoothness.
Physical stop / backend feedback.
Actual robot and camera mounts.

Repository validation

Full CI passed: 696 Python (9 skips), 127 launcher, 687 Specs, 284 WebXR, 49 browser tests. Native compile passed.

Native Preview scenarios remain pending: the external CLAD/MCP installation is unavailable.
Native agent turns cannot be forcibly cancelled; a running turn may issue later skills after a software stop.
Manual admission waits for confirmed turn/navigation/following quiescence. Validate hardware stop behavior.
Native goal-monitor races can still stop a newer goal; source-correlated transport cannot repair that algorithm.
No finite test suite proves every scenario. Physical safety, alignment and visual latency need device evidence.

Implementation map

dimos/ar/replication | ClientCore/replication | both host replication folders | PROTOCOL.md